

INCIDENT INVESTIGATION REPORT

Phishing Email Analysis

Greenholt PLC: Suspicious Customer Email

TryHackMe Lab: Greenholt Phish (SOC / Email Analysis)

Case ID	[IR-2026-001]
Analyst	Shubh
Date of Analysis	[20/09/2026]
Report Status	[Final]
Severity	[Medium]
Verdict	[Phishing]
Classification	[Business Email Compromise attempt with malicious attachment]

Prepared for learning and documentation purposes. This report is based on a training scenario and contains no real customer or company data.

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Document Control

Version	Date	Author	Description
0.1	[20/09/2026]	Shubh	Initial draft
1.0	[20/09/2026]	Shubh	[Final report]

1. Executive Summary

Write this section last. Keep it to 4 to 6 sentences that a non-technical manager could read on its own.

A sales executive at Greenholt PLC reported a suspicious email that appeared to come from a known customer. The email contained a generic greeting, an unexpected request for a money transfer, and an unsolicited attachment. The message was escalated to the SOC for analysis.

Analysis of the email headers, sender infrastructure, and attachment determined that the email is **phishing**. Key findings include **sender domain mismatch**, **malicious attachment hash**.

Recommended actions include **block sender**, **remove email from mailboxes**, **confirm attachment was not opened**, **user awareness**.

2. Background and Scope

2.1 Incident Description

Reported by	Sales executive, Greenholt PLC
Reported via	Employee report to SOC
Claimed sender	Known customer
Reason for suspicion	Generic greeting, unexpected money transfer request, unsolicited attachment, out-of-character communication style

2.2 Objectives

- Analyze the provided email and extract key artifacts
- Investigate the message source to determine its origin and authenticity
- Use analysis tools to assess the potential maliciousness of the email

2.3 Scope

In scope: the submitted email sample (headers, body, attachment) and related sender infrastructure.

Out of scope: endpoint forensics, mailbox-wide hunting, and network traffic analysis, as no such data was provided.

2.4 Assumptions and Limitations

- The analysis relies solely on the provided email sample.
- Attachment analysis was performed **statically** in a controlled environment.

3. Tools and Environment

Tool	Purpose
Text editor / email client	Viewing the raw .eml file
Email header analyzer	Parsing routing path and authentication results
sha256sum / CyberChef	Hashing the attachment
VirusTotal	File, URL, and IP reputation
WHOIS / IP lookup	Ownership and registration data for IPs and domains
DNS lookup (dig / nslookup)	Checking SPF, DKIM, and DMARC records

4. Evidence Handling

Evidence	Filename	SHA-256	Source
Email sample	challenge.eml	NA	Provided by TryHackMe
Attachment	SWT_#09674321____PDF__.CAB	2e91c533615a9bb8929ac4bb76707b2444597ce063d84a4b33525e25074fff3f	Extracted from email

Handling note: The attachment was never opened on a production or personal system. All analysis was performed using hashes and static inspection.

5. Investigation Methodology

The investigation followed a standard phishing triage workflow:

1. **Artifact extraction:** sender, recipient, subject, dates, attachment details
2. **Header analysis:** routing path, originating IP, authentication results
3. **Sender verification:** display name, From, Reply-To, Return-Path consistency
4. **Content analysis:** language, social engineering cues, links
5. **Attachment analysis:** file type, hashing, reputation lookup
6. **Correlation and verdict:** combining the evidence into a conclusion

6. Detailed Analysis

6.1 Email Artifacts

Subject	webmaster@redacted.org your: Transfer Reference Number:(09674321)
Date / Time (with timezone)	6/10/20, 05:58
From (display name)	Mr. James Jackson
From (address)	<info@mutawamarine.com>
Reply-To	Mr. James Jackson
Return-Path	<info.mutawamarine@mail.com>
To	Webmaster
Message-ID	webmaster@redacted.org
Attachment	SWT_#09674321____PDF____.CAB

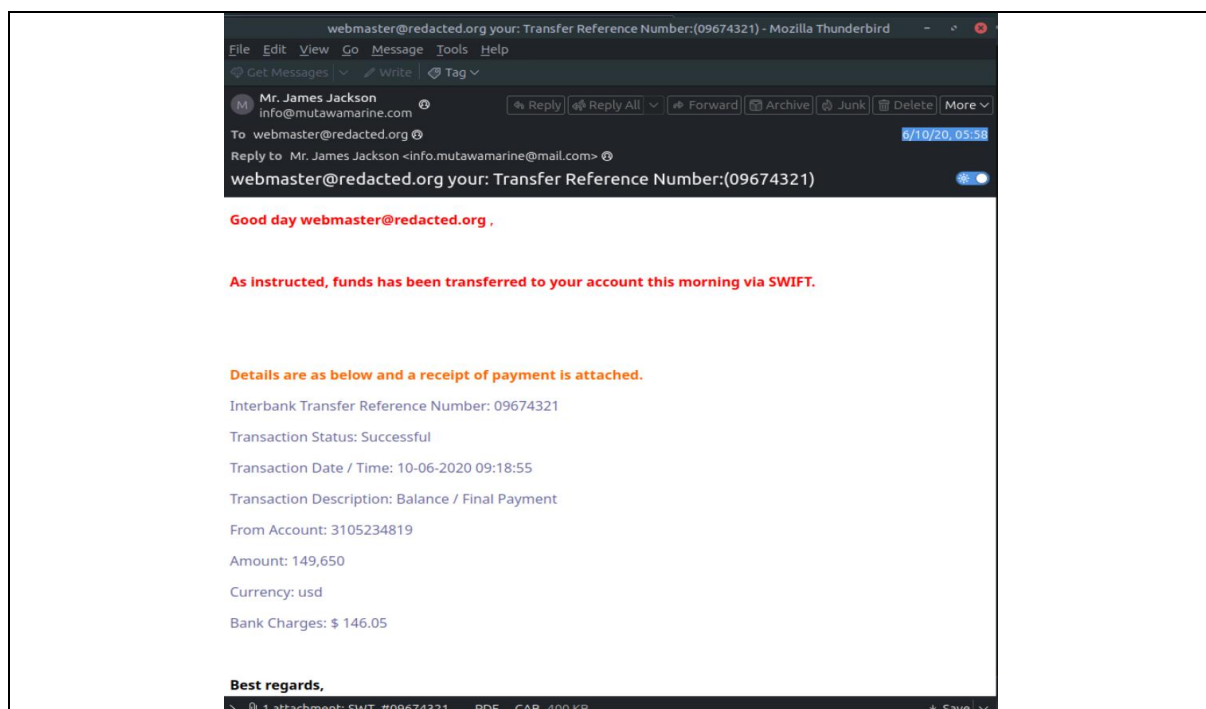


Figure 1: Email overview

Observations:

- The email is timestamped 05:58, but the body claims the transfer happened at 09:18:55 "this morning". If both are in the same timezone, the email predates the supposed transaction. Check the header timezone before you cite this.
- The display name looks legitimate, but the Reply-To sends any response to a different, free mailbox.

6.2 Header and Routing Analysis

Originating IP	192.119.71.157
Sending mail server	HostPapa

6.3 Sender Authentication

Mechanism	Result	Meaning
SPF	Fail	Sending IP was not authorised by domain
DKIM	None	Message was not cryptographically typed
DMARC	Quarantine	No authentication

6.4 Sender Infrastructure Investigation

IP lookup

IP address	192.119.71.157
Owner / ISP	HostPapa
Country	United States

MY IP

IP LOOKUP

HIDE MY IP

VPNS ▾

IP Details For: 192.119.71.157

Decimal: 3229042589

Hostname: client-192-119-71-157.hostwindsdns.com

ASN: 54290

ISP: HostPapa

Services: Data Center/Transit

Country: United States

State/Region: New York

City: Buffalo

Latitude: 42.8868 (42° 53' 12.35" N)

Longitude: -78.8787 (78° 52' 43.24" W)

Toronto

Hamilton

Rochester

Syracuse

Ithaca

Cleveland

Youngstown

Canton

Lake Erie

PENNSYLVANIA

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CLICK TO CHECK BLACKLIST STATUS

Latitude and Longitude are often near the center of population. These values are not precise enough to be used to identify a specific address, individual, or for legal purposes. IP data from [IP2Location](#).

Figure 4: WHOIS and IP lookup results

6.5 Sender Consistency Check

Check	Result	Suspicious?
Display name matches known customer	Yes	yes
From domain matches customer's real domain	No	Yes
Reply-To matches From	yes	Yes
Return-Path matches From	No	Yes

VirusTotal result

Detection ratio	49/64
Detected family / label	Msil, loki
Threat category	Trojan, Ransomware

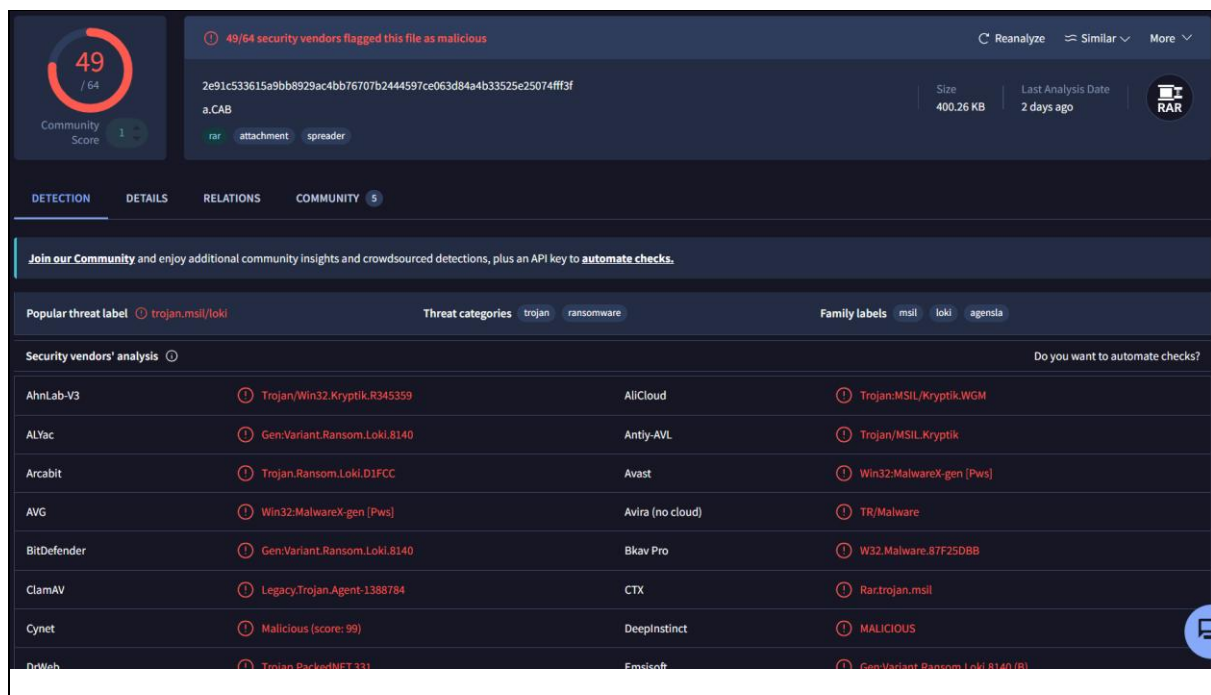


Figure 6: VirusTotal analysis of the attachment

Observations: File was found malicious !

7. MITRE ATT&CK Mapping

Keep only techniques you have evidence for.

Tactic	Technique	ID	Evidence
Initial Access	Phishing: Spearphishing Attachment	T1566.001	Attachment
Defense Evasion	Masquerading	T1036	Extension Spoofing

8. Indicators of Compromise (IOCs)

Defang all values (hxxp, [.), [@]) so they cannot be clicked accidentally.

Type	Indicator	Context
Sender email	info@mutawamarine.com	From address
Reply-To email	info.mutawamarine@mail.com	Reply-To address
Originating IP	192.119.71.157	First external hop
File name	SWT_#09674321__PDF__.CAB	Attachment
SHA-256	2e91c533615a9bb8929ac4bb76707b2444597ce063d84a4b33525e25074fff3f	Attachment

9. Risk Assessment

Likelihood of user interaction	Medium
Potential impact	financial loss
Sophistication	Medium
Targeting	targeted
Overall severity	High

10. Verdict and Conclusion

Verdict	PHISHING
Confidence	High

Conclusion: The mail was found malicious with a malicious attachment intended to create urgency and make a financial loss.

12. Response and Remediation

12.1 Immediate Actions (Containment)

- ☐ Quarantine and delete the email from all mailboxes
- ☐ Block the sender address and domain at the email gateway
- ☐ Block the originating IP and any malicious URLs
- ☐ Add the attachment hash to the endpoint blocklist
- ☐ Confirm with the reporting employee that the attachment was not opened and no payment was made

12.2 If the Attachment Was Opened

- ☐ Isolate the affected endpoint
- ☐ Run a full EDR and antivirus scan
- ☐ Reset the user's credentials
- ☐ Review authentication logs for suspicious sign-ins

12.3 Follow-up Actions

- ☐ Notify the real customer that their identity was impersonated (if applicable)
- ☐ Search for similar emails from the same sender, IP, or attachment hash
- ☐ Report the phishing domain to the registrar and relevant abuse contacts

12.4 Preventive Recommendations

Recommendation	Benefit
Enforce SPF, DKIM, and DMARC with a quarantine or reject policy	Blocks spoofed senders
Flag external emails with a warning banner	Alerts users to outside senders
Require out-of-band verification (call-back) for payment requests	Stops BEC fraud
Sandbox attachments at the mail gateway	Detects malicious files
Run regular phishing awareness training	Improves user reporting

End of Report